What's New

Lab Content

Setting up a Data Lake on AWS

SPL-BE-200-ANDED1-1 - Version 1.0.6

Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

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Lab overview

Example Corp. e-commerce store has a high amount of cart abandonment on a daily basis, and they end up deleting these records from their database to save storage space. As the company’s Data Engineering specialist, you are responsible for finding a cost-effective storage solution in AWS to store these records and allow the e-commerce store to perform analytic processing directly on this storage solution.

In this lab, you explore the components of a data lake, organize your data into layers (or zones), and use Amazon S3 as the storage layer of your data lake.

Objectives

By the end of this lab, you should be able to do the following:

- Use Amazon S3 as the storage layer of a data lake.
- Organize data into layers (or zones) in Amazon S3.
- Configure an S3 event notification to invoke an AWS Lambda function.
- Create an Amazon EventBridge rule to invoke the Lambda function.

Technical knowledge prerequisites

This lab requires the following prerequisites:

- Access to a computer with Microsoft Windows, Mac OS X, or Linux (Ubuntu, SuSE, or Red Hat)
- A modern internet browser such as Chrome or Firefox
- Basic familiarity with cloud and AWS services

Icon key

Various icons are used throughout this lab to call attention to different types of instructions and notes. The following list explains the purpose for each icon:

- **Expected output:** A sample output that you can use to verify the output of a command or edited file.
- **Note:** A hint, tip, or important guidance.
- **Learn more:** Where to find more information.
- **File contents:** A code block that displays the contents of a script or file that you need to run, which has been pre-created for you.
- **Refresh:** A time when you might need to refresh a web browser page or list to show new information.
- **Copy edit:** A time when copying a command, script, or other text to a text editor (to edit specific variables within it) might be easier than editing directly in the command line or terminal.
- **Task complete:** A conclusion or summary point in the lab.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

   Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose

   Open Console

   You are automatically signed in to the AWS Management Console in a new web browser tab.

   Warning: Do not change the **Region** unless instructed.

Common sign-in errors

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the **Start Lab** button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

Lab environment

The following diagram shows the basic architecture of the lab environment:

The architecture diagram of the lab environment.

Image description: The preceding diagram demonstrates a simplified data lake architecture. It contains a layer for raw data (raw zone), a layer for consumable data (consumption zone), and three distinct applications that ingest, process, and consume the data.

The following list details the major resources in the diagram:

- This solution demonstrates a simplified data lake architecture. It contains a layer for raw data (raw zone), a layer for consumable data (consumption zone), and three distinct applications that ingest, process, and consume the data.
- The e-commerce backend application is an AWS Lambda function that ingests data into the raw zone bucket in Amazon S3.
- After the raw data is ingested into the data lake, the processing layer is ready to start transforming the data and sending it to the consumption zone S3 bucket. This bucket contains the transformed data, ready to be consumed.
- A data lake typically has at least three layers (or zones) of data: raw, staging, and consumption. These names can vary, and you can also add additional layers according to your needs.
- When the data is ready to be consumed, the promotions application accesses the data directly in the data lake and aggregates it further to provide abandonment data for each customer. This data can be used to identify which product discounts to send to customers.

Services used in this lab

AWS Lambda

AWS Lambda is a compute service that lets you run code without provisioning or managing servers. Lambda runs your code on a high-availability compute infrastructure and performs all of the administration of the compute resources, including server and operating system maintenance, capacity provisioning and automatic scaling, and logging. With Lambda, all you need to do is supply your code in one of the language runtimes that Lambda supports.

Learn more: Refer to *What is AWS Lambda?* in the **Additional resources** section for more information.

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. Customers of all sizes and industries can use Amazon S3 to store and protect any amount of data for a range of use cases, such as data lakes, websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics. Amazon S3 provides management features so that you can optimize, organize, and configure access to your data to meet your specific business, organizational, and compliance requirements.

Learn more: Refer to *What is Amazon S3?* in the **Additional resources** section for more information.

AWS services not used in this lab

AWS service capabilities used in this lab are limited to what the lab requires. Expect errors when accessing other services or performing actions beyond those provided in this lab guide.

Task 1: Review S3 buckets for raw zone and consumption zone

In this task, you review the S3 buckets that are pre-created for this lab.

3. At the top of the AWS Management Console, in the search bar, search for and choose S3
.

In the **General purpose buckets** section, you should see the following two buckets that are already created for this lab:

- First bucket starting with name **raw-bucket-** is used as a layer for raw data (raw zone).
- Second bucket starting with name **consume-bucket-** is used as a layer for consumable data (consumption zone).

The raw zone (or layer) contains data ingested from the data sources in the raw data format, which is the immutable copy of the data. This zone can include structured, semi-structured, and unstructured data objects such as databases, backups, archives, images, and files (JSON, CSV, XML, text, and so on).

After you have the raw data, you want to transform it. Transformations might involve aggregating the data from different sources or changing the file format of the data received. In this lab, the consumption zone S3 bucket represents the transformed layer.

Task complete: You have successfully reviewed the S3 buckets for raw zone and consumption zone that are pre-created for this lab.

Task 2: Create S3 event notification and send events to Amazon EventBridge

In this task, you create an S3 event notification for the raw zone bucket. You use the Amazon S3 Event Notifications feature to receive notifications when certain events happen in your S3 bucket. Later, you also send these events to Amazon EventBridge.

Task 2.1: Create S3 event notification

4. In the **General purpose buckets** section, choose the raw zone S3 bucket (bucket starting with name **raw-bucket-**).

5. On the raw zone bucket page, choose the **Properties** tab.

6. Scroll down to the **Event notifications** section and choose **Create event notification**.

7. On the **Create event notification** page, in the **General configuration** section, configure the following:

- For **Event name**, enter
  Processor Event
  .

8. In the **Event types** section, for **Object creation**, choose **Put**.

9. Scroll down to the **Destination** section, and configure the following:

- For **Destination**, choose **Lambda function** if it is not already selected.
- For **Specify Lambda function**, choose **Choose from your Lambda functions** if it is not already selected.
- For **Lambda function**, use the dropdown menu and choose **labFunction-Data-Processor**.

In this case, the **labFunction-Data-Processor** Lambda function is selected as the destinations where you want Amazon S3 to send the notifications. You store this configuration in the notification sub-resource that's associated with a bucket.

10. Choose **Save changes**.

    A banner is displayed with the message **Successfully created event notification "Processor Event".** Operation successfully completed.

Task 2.2: Send events to Amazon EventBridge

Amazon S3 can send events to Amazon EventBridge whenever certain events happen in your bucket. Unlike other destinations, you don't need to select which event types you want to deliver.

11. Scroll down to the **Event notifications** section, and for **Amazon EventBridge**, choose **Edit**.

12. On the **Edit Amazon EventBridge** page, for **Send notifications to Amazon EventBridge for all events in this bucket**, choose **On**.

13. Choose **Save changes**.

    A banner is displayed with the message **Successfully edited event notifications.** Operation successfully completed.

Task complete: You have successfully created an S3 event notification for the raw zone bucket, and enabled an option to send events to Amazon EventBridge.

Task 3: Review the ingestion layer for your data lake solution

In this task, you review the ingestion layer for your data lake solution. The **labFunction-Data-Generator** Lambda function acts as the e-commerce backend application that ingests data into the raw zone bucket in Amazon S3.

Task 3.1: Configure and test the labFunction-Data-Generator Lambda function

14. At the top of the AWS Management Console, in the search bar, search for and choose

    Lambda

    .

15. In the **Functions** section, choose the **labFunction-Data-Generator** function.

16. Scroll down to the **Code** tab.

17. **File contents:** In the **index.py** window, review the **labFunction-Data-Generator** function code.

Note: This function creates random shopping cart abandonment data using the Faker Python package.

Learn more: Faker is a python package that generates fake data for you. Refer to *Welcome to Faker's documentation!* in the **Additional resources** section for more information.

18. Choose the **Configuration** tab.

19. On the **Configuration** tab, choose **Environment variables**.

20. On the **Environment variables** section, choose **Edit**.

21. In the **Environment variables** section, for the *input_bucket* **Key**, under **Value**, replace the generic text *REPLACE_WITH_INPUT_BUCKET* with the value of **RawBucketName** provided to the left of these instructions.

22. Choose **Save**.

    A banner is displayed with the message Successfully updated the function labFunction-Data-Generator.

23. Choose the **Test** tab.

24. For **Event name**, enter

TestEvent
.

25. Choose Save.

    A banner is displayed with the message The test event “TestEvent” was successfully saved.

26. To run the test event, choose Test.

27. When the test finishes running, under **Executing function: succeeded**, choose **Details** to view the output.

Expected output: The first five lines of shopping cart data, created by this function, should be displayed.

```
************************
**** EXAMPLE OUTPUT ****
************************

START RequestId: 85ed1153-2c0e-4464-b15e-bfb7fc9bd027 Version: $LATEST
cart_id  customer_id  product_id  product_amount product_price
0       10           3           5              15          $9.47
1        5           0           6              13     $17,048.51
2        0           8          10               9          $5.09
3       10           2           2               1         $25.84
4        0           5           8               9        $690.31
END RequestId: 85ed1153-2c0e-4464-b15e-bfb7fc9bd027
REPORT RequestId: 85ed1153-2c0e-4464-b15e-bfb7fc9bd027  Duration: 4266.55 m

Request ID: 85ed1153-2c0e-4464-b15e-bfb7fc9bd027
```

Task 3.2: Review data in the raw zone S3 bucket

28. At the top of the AWS Management Console, in the search bar, search for and choose

    S3
    .

29. In the **General purpose buckets** section, choose the raw zone S3 bucket (bucket starting with name **raw-bucket-**).

30. On the **Objects** tab, choose the check box to select the **cart_abandonment_data.csv** file.

Refresh: If the .csv file is not displayed, choose the refresh icon.

31. Choose Download, and save the **cart_abandonment_data.csv** file to your device.

32. On your device, open the **cart_abandonment_data.csv** file using an editor of your choice.

Expected output: Notice that the first few entries match what was displayed in the **labFunction-Data-Generator** function test event output.

Task complete: You have successfully reviewed the ingestion layer for your data lake solution.

Task 4: Review the processing layer for your data lake solution

In this task, you review the processing layer for your data lake solution. After the raw data is ingested into the data lake, the processing layer is ready to start transforming the data and sending it to the consumption zone S3 bucket. The **labFunction-Data-Processor** Lambda function acts as the application that transforms the data and sends it to the the consumption zone S3 bucket.

Task 4.1: Configure the labFunction-Data-Processor Lambda function

33. At the top of the AWS Management Console, in the search bar, search for and choose

    Lambda

    .

34. In the **Functions** section, choose the **labFunction-Data-Processor** function.

35. In the **Function overview** section, review to ensure that S3 is listed as the function trigger.

Note: This trigger was added when you configured the event notifications on the raw zone S3 bucket.

36. Scroll down to the **Code** tab.

37. **File contents:** In the **index.py** window, review the **labFunction-Data-Processor** function code.

Note: This function aggregates the shopping cart abandonment data and sorts it by product ID, using the pandas Python data analysis library.

Learn more: Pandas is a fast, powerful, and flexible open-source data analysis and manipulation tool, built on top of the Python programming language. Refer to *pandas* in the **Additional resources** section for more information.

38. Choose the **Configuration** tab.

39. On the **Configuration** tab, choose **Environment variables**.

40. On the **Environment variables** section, choose **Edit**.

41. In the **Environment variables** section:

- For the *input_bucket* **Key**, under **Value**, replace the generic text *REPLACE_WITH_INPUT_BUCKET* with the value of **RawBucketName** provided to the left of these instructions.
- For the *output_bucket* **Key**, under **Value**, replace the generic text *REPLACE_WITH_OUTPUT_BUCKET* with the value of **ConsumeBucketName** provided to the left of these instructions.

42. Choose **Save**.

    A banner is displayed with the message Successfully updated the function labFunction-Data-Processor.

43. On the top breadcrumb menu, choose **Functions**.

44. In the **Functions** section, choose the **labFunction-Data-Generator** function.

45. Scroll down to the **Code** tab.

46. To run the test event a second time. Choose the Test tab and choose Test.

Expected output:

```
************************
**** EXAMPLE OUTPUT ****
************************

Status: Succeeded
Test Event Name: TestEvent

Response:
null

The area below shows the last 4 KB of the execution log.

Function Logs:
START RequestId: b6cfd7bf-8e13-4ac6-bc29-739cf4dcca8f Version: $LATEST
cart_id  customer_id  product_id  product_amount product_price
0       10            1           9              4     $8,295.85
1        8            3          10             17     $3,482.04
2       10            3           7             18       $788.59
3        4            6           9              6     $2,936.20
4        6            2           3              4         $9.76
END RequestId: b6cfd7bf-8e13-4ac6-bc29-739cf4dcca8f
REPORT RequestId: b6cfd7bf-8e13-4ac6-bc29-739cf4dcca8f  Duration: 1305.71 m:

Request ID: b6cfd7bf-8e13-4ac6-bc29-739cf4dcca8f
```

47. In the **Output** tab, review the output to ensure that the test event completed successfully.

Task 4.2: Review data in the consumption zone S3 bucket

48. At the top of the AWS Management Console, in the search bar, search for and choose

 S3

.

49. In the **General purpose buckets** section, choose the consumption zone S3 bucket (bucket starting with name **consume-bucket-**).

50. On the **Objects** tab, choose the check box to select the **cart_aggregated_data.csv** file.

Refresh: If the .csv file is not displayed, choose the refresh icon.

51. Choose **Download**, and save the **cart_aggregated_data.csv** file to your device.

52. On your device, open the **cart_aggregated_data.csv** file using an editor of your choice.

Note: The **labFunction-Data-Processor** function has aggregated the total amount of abandoned products sorted by product ID.

Task complete: You have successfully reviewed the processing layer for your data lake solution.

Task 5: Review the consumption layer for your data lake solution

In this task, you review the consumption layer for your data lake solution.

Now as the data is ready to be consumed, the **labFunction-Promotion-App** Lambda function acts as the promotions application that accesses the data directly in the data lake and aggregates it further to provide abandonment data for each customer. This data can be used to identify which product discounts to send to customers.

Task 5.1: Create an Amazon EventBridge rule

53. At the top of the AWS Management Console, in the search bar, search for and choose
 Amazon EventBridge

.

Learn more: Amazon EventBridge is a serverless event bus service that you can use to connect your applications with data from a variety of sources. Refer to *What Is Amazon EventBridge?* in the **Additional resources** section for more information.

54. On the **Amazon EventBridge** console home page, choose **Create rule**.

Note: On top of the page, if you see a banner for **Visual rule builder opt in**, choose the toggle button to disable it.

55. In the **Define rule detail** step, in the **Rule detail** section, configure the following:

- For **Name**, enter
  PromotionApp
  .

56. Choose **Next**.

Note: A rule matches incoming events and sends them to targets for processing. A single rule can send an event to multiple targets, which then run in parallel. Rules are based either on an event pattern or a schedule. An event pattern defines the event structure and the fields that a rule matches.

57. In the **Build event pattern** step, scroll down to **Event pattern** section, and configure the following:

- For **Event source**, choose **AWS services** if it is not already selected.
- For **AWS service**, use the dropdown menu and choose **Simple Storage Service (S3)**.
- For **Event type**, use the dropdown menu and choose **Amazon S3 Event Notification**.
- In the **Event pattern** text box, review that the detail-type JSON structure has multiple event types that should trigger the rule for any bucket.

Expected output: The JSON structure should look similar to the following:

```
************************
**** EXAMPLE OUTPUT ****
************************


{
  "source": ["aws.s3"],
  "detail-type": ["Object Access Tier Changed", "Object ACL Updated", "Obje
}
```

- For **Event Type Specification 1**, choose **Specific event(s)**.
- For **Specific event(s)**, on the dropdown menu, choose **Object Created**.
- For **Event Type Specification 2**, choose **Specific bucket(s) by name**.
- For **Specific bucket(s) by name**, enter the value of **RawBucketName** provided to the left of these instructions.

58. Choose **Next**.

59. In the **Select target(s)** step, in the **Target 1** section, configure the following:

- For **Target types**, choose **AWS service**.
- For **Select a target**, use the dropdown menu, and choose **Lambda function**.

- For **Function**, use the dropdown menu, and choose **labFunction-Promotion-App**.
- Under **Permissions** uncheck **Use execution role (recommended)**

60. Choose **Next**.

61. In the **Configure tags -** ***optional*** step, choose **Next**.

62. In the **Review and create** step, review the **Event pattern** and confirm it looks similar to the following example:

```
**********************
**** EXAMPLE OUTPUT ****
**********************

{
  "source": ["aws.s3"],
  "detail-type": ["Object Created"],
  "detail": {
    "bucket": {
      "name": ["rawbucket-29387"]
    }
  }
}
```

63. Scroll down to the bottom of the page and choose **Create rule**.

    A banner is displayed with the message Rule PromotionApp was created successfully.

Task 5.2: Configure the labFunction-Promotion-App function

64. At the top of the AWS Management Console, in the search bar, search for and choose

    Lambda
    .

65. In the **Functions** section, choose the **labFunction-Promotion-App** function.

66. In the **Function overview** section, review to ensure that **EventBridge (CloudWatch Events)** is listed as the function trigger.

Note: This configuration was added automatically when you set up the **PromotionsApp** EventBridge rule with the **labFunction-Promotion-App** as the target.

67. Scroll down to the **Code** tab.

68. **File contents:** In the **index.py** window, review the **labFunction-Promotion-App** function code.

Note: This function further aggregates and sorts the cart abandonment data by customer ID to get the top abandoned products for each customer.

69. Choose the **Configuration** tab.

70. On the **Configuration** tab, choose **Environment variables**.

71. On the **Environment variables** section, choose **Edit**.

72. In the **Environment variables** section:

- For the *input_bucket* **Key**, under **Value**, replace the generic text *REPLACE_WITH_INPUT_BUCKET* with the value of **RawBucketName** provided to the left of these instructions.
- For the *output_bucket* **Key**, under **Value**, replace the generic text *REPLACE_WITH_OUTPUT_BUCKET* with the value of **ConsumeBucketName** provided to the left of these instructions.

73. Choose **Save**.

    A banner is displayed with the message Successfully updated the function labFunction-Promotion-App.

74. In the **Functions** section, choose the **labFunction-Data-Generator** function.

75. Scroll down to the **Code** tab.

76. To run the test event a second time. Choose the Test tab and choose Test.

Expected output:

```
************************
**** EXAMPLE OUTPUT ****
************************

Status: Succeeded
Test Event Name: TestEvent

Response:
null

The area below shows the last 4 KB of the execution log.

Function Logs
START RequestId: 1a674776-37a2-4c61-bee3-4a6b3917fdfb Version: $LATEST
cart_id  customer_id  product_id  product_amount product_price
0        9            2           7              18       $845.86
1        5           10          10               7       $673.57
2        9            2           1              19        $27.18
3        3            3           3               4         $4.48
4        9            8           7              10    $85,998.01
```

```
END RequestId: 1a674776-37a2-4c61-bee3-4a6b3917fdfb
REPORT RequestId: 77eb838c-d774-4848-b9b9-d6cead19d653  Duration: 5070.66 m:

Request ID: 1a674776-37a2-4c61-bee3-4a6b3917fdfb
```

77. In the **Output** tab, review the output to ensure that the test event completed successfully.

Task 5.3: Review data in the consumption zone S3 bucket

78. At the top of the AWS Management Console, in the search bar, search for and choose

    S3

    .

79. In the **General purpose buckets** section, choose the consumption zone S3 bucket (bucket starting with name **consume-bucket-**).

80. On the **Objects** tab, choose the check box to select the **promotion_data.csv** file.

Refresh: If the .csv file is not displayed, choose the refresh icon.

81. Choose Download, and save the **promotion_data.csv** file to your device.

82. On your device, open the **promotion_data.csv** file using an editor of your choice.

Note: The **labFunction-Promotion-App** function aggregated the cart abandonment data and sorted it by customer ID. The company can use this data to identify the top few products that customers leave in shopping carts, and then offer promotional discounts for these products.

Task complete: You have successfully reviewed the consumption layer for your data lake solution.

Conclusion

You have successfully done the following:

- Used Amazon S3 as the storage layer of a data lake.
- Organized data into layers (or zones) in Amazon S3.
- Configured an S3 event notification to invoke an AWS Lambda function.
- Created an Amazon EventBridge rule to invoke the Lambda function.

End lab

Follow these steps to close the console and end your lab.

83. Return to the **AWS Management Console**.

84. At the upper-right corner of the page, choose **AWSLabsUser**, and then choose **Sign out**.

85. Choose **End Lab** and then confirm that you want to end your lab.

Additional resources

- [What is AWS Lambda?]
- [What is Amazon S3?]
- [Welcome to Faker’s documentation!]
- [pandas]
- [What Is Amazon EventBridge?]

For more information about AWS Training and Certification, see *https://aws.amazon.com/training/*.

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Nice work! Help us make Builder Labs better in just 2 minutes.